



fluxcore

TWO-DAY SPRINT PLAN // VIRTUAL POWER PLANT ARBITRAGE ENGINE

Ship a live, demoable VPP trading desk — real ERCOT prices, real strategies, real money left on the table — in two days, with strict TDD the whole way down.

SPRINT GOAL

A stranger with a browser can watch a 200-home battery fleet trade the ERCOT real-time market — live, replayed, and backtested against a perfect-hindsight oracle — at fluxcore.solutions.

CEO

Michael

Product vision and the pitch. Owned the demo narrative and run-of-show, framed the VPP arbitrage story for the room, and ran acceptance passes on the live site as the voice of the audience.

CTO

Penn

Architecture and engineering direction. Set the strict-TDD mandate, drove the simulation core, live spine, backtest lab + oracle, the Cloudflare stack (Pages, Worker, DO, D1, KV), CI, integration, and release ops.

COO

Diana

Operations and delivery. Owned the fluxcore.solutions domain and DNS cutover, the brand assets, and shipped the 9-PR feature track: fleet designer, Monte Carlo stress, wear dashboard, export, storm mode, hotkeys.

SPRINT WINDOW

2026-06-10 → 06-12

DEMO

2026-06-12

REPO

[melonmelonz/fluxcore](https://github.com/melonmelonz/fluxcore)

LIVE AT

fluxcore.solutions

PRIORITIES Three pillars, in order

1 · Correct physics & money

Battery, solar, fleet, ledger, strategies: pure, deterministic, unit-tested core. One **deskTick** executor feeds replay, live desk, and backtest — one set of physics, no drift.

2 · Real data, owned forever

Zero paid or capped APIs at runtime. ERCOT prices scraped from public reports and write-through archived into D1 — the dataset grows itself every 15 minutes.

3 · A narrative the room can feel

"The threshold bot made \$22k. The optimizer made \$17k. The oracle proves \$25.7k was available. Here is the gap — and here is why."

EPIC 1 Simulation Core

DONE · DAY 1 AM

Strict red/green TDD for every module. Logs in docs/tdd/logs/, rendered in the README.

#	STORY	TDD LOG	COMMIT
1.1	Scenario types + boundary parser	02	b81e091
1.2	Battery state machine — rate / capacity / efficiency clamping	03	3a2e0b7
1.3	Clear-sky solar generation profile	04	c0215cf
1.4	Fleet aggregation, proportional dispatch, free solar charging	05	87314ca
1.5	Market clock replay stepper	06	8f3b2d9
1.6	Threshold baseline strategy with breakeven guard	07	790d761
1.7	LP optimizer strategy over day-ahead horizon	08	0d5903a
1.8	Dispatch ledger with running P&L	09	0733344
1.9	Simulation controller with per-strategy lanes	10	e8f0dae

Acceptance (met): deterministic replay of the 2023 Texas heatwave week — threshold **\$22,116.69**, LP **\$17,516.79**, byte-for-byte reproducible.

EPIC 2 Data, UI, Brand, CI

DONE · DAY 1 PM

#	STORY	EVIDENCE
2.1	Ingest script + bundled real ERCOT scenarios	20f26e5 · log 11
2.2	App shell, design tokens, simulation hook	4975433
2.3	Dashboard: live chart, dispatch markers, lane cards	ed272ef · log 13
2.4	Brand: logo, favicon, vivid blue/cyan + orange palette	302688f b5b413d
2.5	Strict security headers + eslint	2787d87
2.6	CI: test gate + Pages deploy on main	6b9f4fd
2.7	README with red/green screenshots for every module	1e6cba9

EPIC 3 Plan A — The Live Spine

DONE · DAY 2 AM

Constraint honored: **zero gridstatus.io calls at runtime** (free-tier caps). ERCOT MIS public reports scraped directly; every point fetched is archived forever (Epic 6).

#	STORY	TDD LOG	COMMIT
3.1	ERCOT MIS report parser, real-report zip fixtures	20	866e275
3.2	Rolling price cache merge — dedupe, age trim	21	530e7b2
3.3	Shared deskTick executor + fleet state serialization	22	e0d6cf6
3.4	fluxcore-desk Worker: cron ingest, D1 archive, LiveDesk Durable Object	—	6544b4b
3.5	Same-origin Pages Functions API over KV + DO bindings	—	1598938
3.6	Live mode UI: desk polling hook, freshness badge	23	f8b756a
3.7	CI: worker/functions typecheck; worker deploy (token scopes pending)	—	3b952a2

Architecture note: CF Pages cannot declare Durable Objects inline — the DO lives in a sibling Worker, bound via script_name. Lesson institutionalized.

EPIC 4 Plan B — Backtest Lab + Oracle

DONE · DAY 2 PM

#	STORY	TDD LOG	COMMIT
4.1	Historical archive: 96 monthly JSON chunks, 4 hubs, 2023–2024 (~12 MB, static)	—	19a9fbb
4.2	Perfect-hindsight oracle: 48h receding-horizon LP, SoC carryover, curtailable solar	25	ce9ee7c
4.3	Steppable max-speed backtest runner + “never beats the oracle” property test	26	b7eae0
4.4	Lab range helpers + shareable URL codec <code>#lab?hub=..&start=..&end=..</code>	27	c553ce3
4.5	Lab UI: hub/date pickers, chunked progress, oracle table, “left on the table”	—	a75bf34
4.6	Deploy, screenshots, README “live spine and the lab” chapter	—	e1de188

Design call worth remembering: a day-chunked oracle LP provably fails overnight carry — it has no motive to charge at 23:45 for tomorrow’s spike. The fix (solve a 48h window, commit only day one, carry SoC) fell straight out of a failing test. **TDD earned its keep here.**

Acceptance (met): heatwave week in the Lab — threshold **\$22,093.63** (85.8% of oracle), LP **\$17,516.79** (68.0%), oracle **\$25,748.79**. Share links cold-load and auto-run.

EPIC 5 Production Hardening — The Imperva Campaign

PARTIAL · ONGOING

ERCOT fronts MIS with Imperva. It serves Workers egress an HTML JS-challenge; residential IPs pass clean.

#	STORY	STATUS	EVIDENCE
5.1	Pick newest <code>_csv</code> variant from doc list (defensive)	DONE	log 24 · 959075b
5.2	Cookie-jar retry helper: accumulating jar, browser headers, KV-persisted cookies	DONE*	log 28 · 02212ac
5.3	Diagnosis: tail capture proves the challenge defeats cookie replay — it needs script execution, not cookie echo	DONE	worker tail evidence
5.4	Egress probe: <code>systemWidePrices.json</code> returns 200 + a full day of 15-min hub SPPs to Workers egress	DONE	probe results
5.5	Switch RTM ingestion to dashboard JSON (parser + fallback chain)	OPEN	top of day-3 backlog

Demo posture meanwhile: the desk fails closed and serves cached prices behind an honest freshness badge; replay + Lab are fully self-contained and carry the demo regardless.

EPIC 6 Self-Growing Archive

DONE

Every price point that ever reaches the worker is INSERT OR REPLACEd into D1. The Lab dataset seeds 2023–2024; the cron grows it forward forever. No API caps, no vendor, no decay.

EPIC 7 Launch Ops — Domain + Team

DONE · DEMO DAY

#	STORY
7.1	Register <code>fluxcore.solutions</code> + <code>www</code> on the Pages project via CF API (wrangler has no subcommand for this)
7.2	CF zone created; GoDaddy parked A records purged; apex CNAME-flattened to <code>fluxcore-30a.pages.dev</code>
7.3	Nameserver cutover GoDaddy → Cloudflare; registry delegation verified at the TLD servers
7.4	Stuck cert validation root-caused (domains registered before the zone existed); delete + re-add forced instant validation
7.5	End-to-end verify: apex + <code>www</code> 200, <code>/api/state</code> 200, archive index 200
7.6	Collaborator audit: both partners confirmed write access

EPIC 8 Diana’s Feature Set — parallel track

DONE · 9 PRs

Spec’d and planned in docs/superpowers, delivered as 9 reviewed PRs (#1–#9), ~2,850 lines.

#	STORY	PRS
8.1	Brand assets: new logo crop, emblem-only vector favicon, header lockup	#1 #8 #9
8.2	Storm mode: spike detection + spike-triggered UI shift with a11y badge	#2
8.3	Battery wear dashboard: cycles, throughput totals, degradation dollars	#3
8.4	One-click CSV/JSON export for dispatch log and lab runs	#4
8.5	Fleet designer (was Plan D): unit catalog, presets, validated mix storage, wired through replay desk AND backtest lab	#5
8.6	Monte Carlo stress card: seeded PRNG, forecast noise, steppable runner, streaming histogram	#6
8.7	Keyboard shortcuts: space play/pause, 1/2/3 speed; inline control bar	#7

Integration (merge day): docs-only rebase, zero conflicts — the shared deskTick/lane architecture absorbed a fleet-mix dimension without touching core physics. One environment fix followed: Node ≥ 25’s stub `localStorage` shadows `jsdom`’s under vitest 4; an in-memory polyfill keeps the suite green everywhere.

SCOREBOARD Sprint Metrics

136

TESTS, 28 FILES, ALL GREEN

20

RED/GREEN TDD LOG PAIRS

70+

COMMITTS ON MAIN

9

PARTNER PRS MERGED, 0 CONFLICTS

96

ARCHIVE CHUNKS · 4 HUBS · 2 YRS

\$25.7k

ORACLE BOUND, HEATWAVE WEEK

\$0

RUNTIME API SPEND

100%

LIVE AT FLUXCORE.SOLUTIONS

RISK LOG As materialized

RISK	HIT?	RESPONSE
gridstatus free-tier caps	avoided	Direct MIS scraping + local pickle ingest; gridstatus only used offline to build the archive
Imperva blocks Workers egress	HIT	Cookie-jar + KV persistence shipped; root-caused as JS challenge; bypass found (dashboard JSON, probe-verified 200); swap queued
CF Pages can't host Durable Objects	hit early	Sibling Worker pattern, bound via <code>script_name</code>
Day-chunked oracle under-bounds	HIT	Caught by test. 48h receding horizon, commit-first-day, SoC carryover
Domain cutover the night before demo	survived	NS delegation watched at the registry; stuck validation fixed by re-registering domains against the active zone
Unicode in commit messages breaks Pages CI	avoided	ASCII-only commit discipline

RUNBOOK Demo Day · 2026-06-12

- 1 Open <https://fluxcore.solutions> — hard-refresh once (a GoDaddy parking service worker may linger in any browser that visited pre-cutover).
- 2 **Act 1 — Replay:** heatwave week; threshold vs LP lanes diverge, dispatch markers on the chart. End on the P&L: \$22.1k vs \$17.5k.
- 3 **Act 2 — Lab:** run the same week against the oracle. Punchline: even the winner left **\$3.6k on the table**.
- 4 **Act 3 — Share:** paste a share link into a fresh tab — cold-loads and auto-runs. “Send your own backtest to anyone.”
- 5 **Act 4 — Live:** the live desk + freshness badge; narrate the self-growing D1 archive, and the Imperva war story if data sourcing comes up.
- 6 Fallback URL if anything regional breaks: fluxcore-30a.pages.dev (identical deployment).

RETRO Two days, honestly

WHAT WENT WELL

- TDD wasn't ceremony — it caught the oracle's overnight-carry bug and the doc-list variant bug before either reached the demo.
- One deskTick for replay/live/backtest meant the Lab shipped in an afternoon with zero physics drift.
- Archive-everything turned an API-caps constraint into a permanent asset.

WHAT WAS HARD

- Imperva. Two escalations before accepting the challenge needs JS execution; an egress probe then found a cleaner door in 10 minutes.
- DNS cutover under time pressure: verify at the registry, not through resolver caches.

DO DIFFERENTLY

- Probe ERCOT's public dashboard endpoints *first*, before building the MIS scrape path.
- Create the CF zone before registering Pages custom domains, so validation never lands in the stale state.